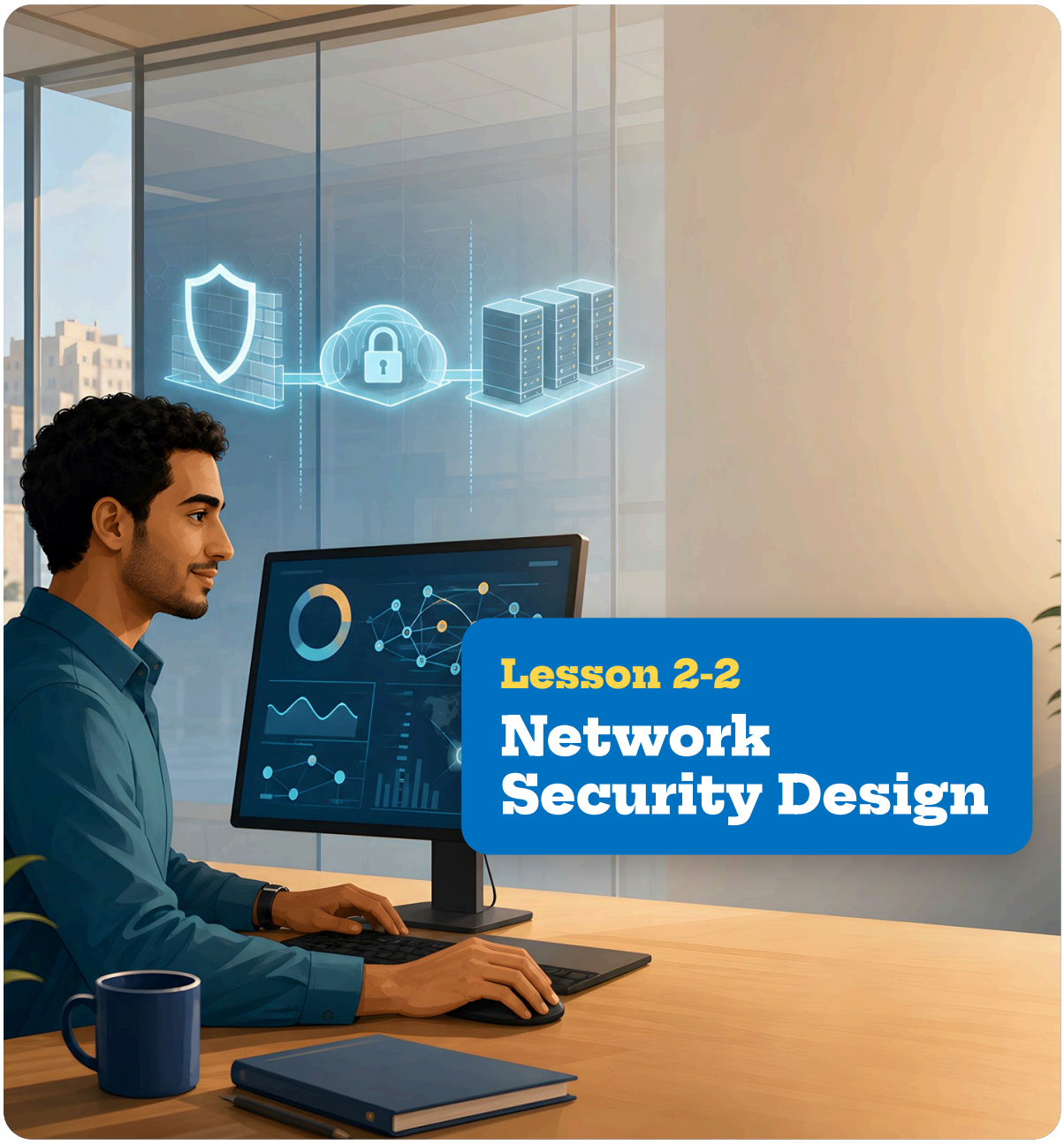


Learning Objectives

- Explain the roles of firewalls, VPNs, and DMZs in protecting a network.
- Describe defense in depth and zero trust, and explain why layered, verified access is stronger than a single barrier.

An organization's network holds confidential data — customer records, financial files, and private messages. Attackers constantly look for a way in. A single lock on the front door is not enough: if it is picked, everything inside is exposed. For this reason, modern network security uses several layers, so that even when one defense is breached, the next one still protects the system.

? Guiding question: How can an organization design its network so that, even if one defense is breached, the system stays protected?

Point!

1. The Role of Firewalls

(1) **(Firewall)**…a system installed at the entry/exit point of a network that allows or denies communication based on rules set in advance.
e.g., Allow access to the web server, but deny direct external access to the database server.

2. VPN (Virtual Private Network)

(1) **(VPN)**…a technology that creates an encrypted virtual private line (tunnel) over the Internet, allowing safe connection to a network from a remote location.

(2) Main uses of VPN

① Remote work…connecting safely to an organization's network from home or while away.

② Connecting branches…connecting offices in different locations safely.

(3) Because communication content is encrypted when using a VPN, the risk of eavesdropping is reduced even when using public Wi-Fi.

3. DMZ (Demilitarized Zone)

(1) **(DMZ)**…an area where servers exposed to the external network (web servers, mail servers, etc.) are placed separately from the internal network.

(2) By providing a DMZ, even if a public-facing server is attacked, damage to the internal network can be prevented.

Key Fact:
Strong network security comes from layering measures — firewall, VPN, DMZ, endpoint protection — and from verifying every access (zero trust), not from a single barrier.

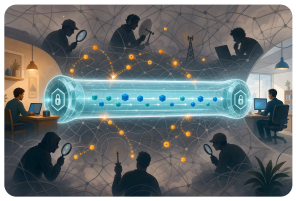
Key Concepts:

- firewall
- VPN
- DMZ
- defense in depth
- zero trust
- perimeter

KEY TERMS
firewall — allows or denies communication at a network's entry/exit point based on preset rules.

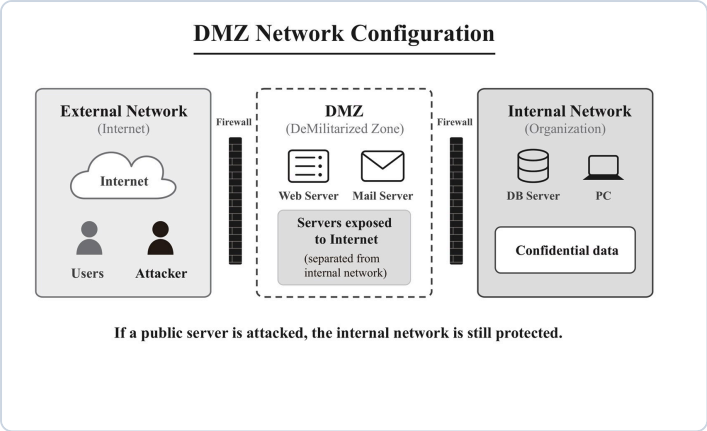


A firewall allows or blocks traffic by rules.



A VPN tunnels data safely through a hostile network.

KEY TERMS
DMZ — a zone for public-facing servers, separated from the internal network.



DMZ Network Configuration

4. Defense in Depth

(1) **(Defense in depth)**…the concept of layering multiple security measures so that, even if one measure is breached, the next can still protect the system.

(2) In network security, multiple measures are combined as follows.

Layer	Examples of measures
Layer 1: Network entry	Block unauthorized communication using a firewall
Layer 2: Communication path	Encrypt communication using a VPN
Layer 3: Server placement	Separate public-facing servers from the internal network using a DMZ
Layer 4: Endpoints	Install antivirus software and keep the OS updated

5. The Concept of Zero Trust

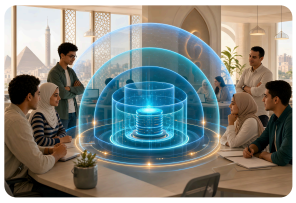
(1) **(Zero trust)**…a security concept that means the system does not trust any communication, including from within the organization's network, and always verifies it.

e.g., Even for access from inside the organization, identity verification and access permission checks are performed every time.

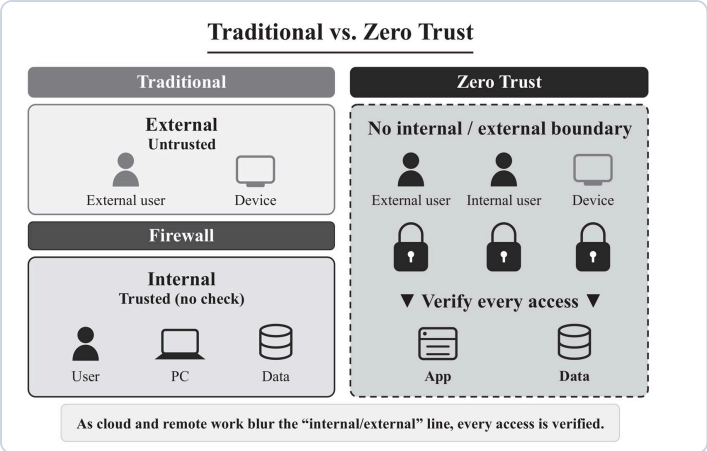
(2) Conventional network security was designed mainly around protecting a perimeter, based on the premise that "inside the organization's network is safe, and outside is dangerous." However, with the spread of cloud services and remote work, the boundary between "inside" and "outside" has become blurred, and designs that protect only the perimeter have become insufficient.

PAUSE & THINK

Why is it safer to place a public web server in a DMZ rather than inside the internal network?



Defense in depth: if one layer falls, the next holds.



Traditional vs. Zero Trust

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A A firewall allows or denies communication based on rules set in advance.

B Using a VPN can reduce the risk of eavesdropping even on public Wi-Fi.

C A DMZ is a system that exposes all servers to the outside.

D Defense in depth is the concept of layering multiple security measures.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a A technology that creates an encrypted virtual private line over the Internet for safe connection from a remote location

b An area where servers exposed to the outside are placed separately from the internal network

c A system that allows or denies communication based on rules set in advance

[Choices] **A** Firewall **B** VPN **C** DMZ

KEY TERMS

defense in depth — layering measures.

zero trust — verify every access.

✓ Solution

(1)

A A firewall controls communication based on rules. Therefore, ○.

B A VPN encrypts communication, reducing the risk of eavesdropping on public Wi-Fi. Therefore, ○.

C A DMZ is an area where public-facing servers are placed separately from the internal network, not a system that exposes all servers. Therefore, ×.

D Correct definition of defense in depth. Therefore, ○.

(2)

a: An encrypted virtual private line is a VPN. **B**

b: An area where public-facing servers are isolated is a DMZ. **C**

c: A system that allows or denies communication is a firewall. **A**

Try

1 Answer the following questions.

1. What is the term for the system installed at the entry/exit point of a network that allows or denies communication based on rules?
2. What is the three-letter abbreviation for the technology that creates an encrypted virtual private line over the Internet for safe communication?
3. What is the three-letter abbreviation for the area where servers exposed to the outside are placed separately from the internal network?
4. What is the term for the concept of layering multiple security measures?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate use of a VPN.

A Connecting safely from home to an organization's network

B Connecting offices in different locations safely

C Automatically deleting files infected with viruses

D Encrypting communication content on public Wi-Fi

(2) For each of the security measures (a–d), indicate which layer of defense in depth (A–D) it belongs to.

a Blocking unauthorized communication with a firewall

b Encrypting communication with a VPN

c Installing antivirus software

d Separating public-facing servers from the internal network using a DMZ

[Layers] **A** Network entry **B** Communication path **C** Server placement **D** Endpoints

Think as an Engineer — Investigate & Decide

A company in your city is setting up its office network. It has a public website, staff who work from home, and confidential customer data. You advise on the security design.

EXAM-STYLE QUESTION

Explain why layering several security measures (defense in depth) protects an organization better than relying on a firewall alone. **[6] Refer to:** what happens if one measure is breached.

☐ A firewall only ☐ **A layered design (firewall + VPN + DMZ + endpoints) that also verifies every access (zero trust)**

1 · Collect data. List the company's three assets (public website, remote staff, customer database) and the main threat to each.

2 · Rank the layers. For this company, rank the four defense-in-depth layers (network entry, communication path, server placement, endpoints) from most to least critical, and justify your top choice.

3 · Decide. Recommend a design and give two reasons why layering is safer than a single firewall. **In pairs,** compare your designs and agree together on the strongest reason before you write it down.

Give evidence for your answer.

Stuck? Start with the most exposed asset — the public website — and place it in a DMZ so that an attack there cannot reach the internal database.

🌍 In a New Context

A hospital moves its patient records to a cloud service and lets doctors access them from home. Using zero trust and defense in depth, suggest two measures it should apply and say where each one helps, then identify one risk that could still remain.

❓ Lesson Question — Answered

The lesson asked how to design a network so that one breach does not expose everything. The answer is layering. A firewall controls what enters and leaves the network; a VPN encrypts communication from remote locations; a DMZ separates public-facing servers so that an attack on them cannot reach internal data; and endpoint measures protect each device. Because cloud services and remote work have blurred the line between "inside" and "outside," zero trust adds a further principle: no access is trusted automatically, and every request is verified. Together, defense in depth and zero trust keep the network protected even when a single measure fails.

KEY TERMS

perimeter — the old "inside is safe" boundary that cloud and remote work have blurred.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

To protect an organization's network, it is important to combine multiple security measures. First, a (a) is installed at the entry/exit point of the network to block unauthorized communication based on rules. Web servers and other servers exposed to the outside are placed in a (b), separated from the internal network. To connect safely from outside, such as with remote work, communication is encrypted using a (c). The concept of layering multiple measures in this way is called (d).

1. Fill in the blanks (a)–(d).

3 Answer the following questions.

(1) From the following options (A–D), choose the most appropriate reason for providing a DMZ.

A To improve communication speed.

B To expose all servers to the outside.

C To prevent damage to the internal network even if a public-facing server is attacked.

D To automatically detect and remove viruses.

(2) From the following options (A–D), choose the one that most appropriately describes the concept of zero trust.

A Access from within the organization's network is safe, so verification is unnecessary.

B A firewall alone is sufficient to protect the network.

C No communication is trusted and is always verified.

D Only access from outside needs to be monitored.

⭐ REFLECT & CHALLENGE

Reflect: Which layer of defense do you think is easiest for an attacker to get past, and why?
Challenge: Suggest one measure a small company could add beyond a firewall, and one threat it would still not remove.

⭐ Key takeaway

Remember:
Layer your defenses — firewall, VPN, DMZ, endpoints — and verify every access (zero trust). One breach is then far less likely to expose the whole network.